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A Project Report on

**LUNGINSIGHT: AN INTELLIGENT LUNG DISEASE DETECTION
AND ANALYSIS SYSTEM**

Submitted in partial fulfillment of the requirements for the conferment of degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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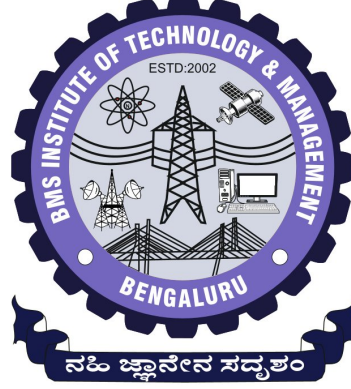
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CERTIFICATE

This is to certify that the project entitled "**LungInsight: An Intelligent Lung Cancer Detection and Analysis System**" is a bonafide work carried out by **Bhoomika S Gowda (1BY23AI029)**, **Devathi Saranya (1BY23AI041)**, **Emma Kappen (1BY23AI049)**, and **Honey Hemant Hampannavar (1BY24AI405)** in partial fulfillment for the award of "**Bachelor of Engineering**" in "**Artificial Intelligence and Machine Learning**" of the **Visvesvaraya Technological University, Belagavi**, during the year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect to work for the B.E. degree.

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ABSTRACT

Lung cancer remains one of the leading causes of cancer-related mortality worldwide, largely due to late-stage detection and reliance on specialist radiologists for CT scan interpretation. LungInsight addresses these challenges by developing a multimodal, explainable AI framework that integrates CT imaging and structured clinical data to enable early and accurate lung cancer prediction. The system employs a DenseNet-based CT image encoder and an MLP-based clinical feature encoder, with late fusion combining both learned representations to jointly predict malignancy probability and disease stage. Training is performed on the LIDC-IDRI dataset paired with National Lung Screening Trial clinical records, using AdamW optimisation with focal loss to handle class imbalance effectively.

Explainability is central to the design: Grad-CAM generates localised heatmaps highlighting suspicious nodule regions in CT slices, while SHAP values quantify the contribution of individual clinical variables such as smoking history, age, and lung function to each prediction. The framework is evaluated using accuracy, ROC-AUC, F1-score, PR-AUC, Brier score, and calibration curves, with ablation studies comparing imaging-only, clinical-only, and fused configurations. A Streamlit-based clinical decision support dashboard enables clinicians to upload CT scans and patient records, run real-time inference, and review explainable risk assessments — making LungInsight a trustworthy, interpretable tool for supporting early lung cancer diagnosis and confident clinical decision-making.

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Declaration

We, hereby declare that the project titled "LungInsight: A Deep Learning Approach for Lung Disease Detection and Classification" is a record of original project phase-2 work undertaken for the award of the degree Bachelor of Engineering in Artificial Intelligence and Machine Learning of the Visvesvaraya Technological University, Belagavi during the year 2025-26. We have completed this project phase -2 work under the guidance of Prof. Megha S, Asst. Professor, Dept. of Artificial Intelligence and Machine Learning.

I also declare that this project report has not been submitted for the award of any degree, diploma, associate ship, fellowship or other title anywhere else

Name	USN	Signature

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Chapter 1

Introduction

1.1 Context and Background

Lung cancer is one of the most prevalent and fatal malignancies worldwide, accounting for a disproportionately high share of cancer-related deaths due to its characteristically late-stage presentation at the time of diagnosis. Early detection remains the most significant determinant of patient survival; however, identifying malignant pulmonary nodules in CT scans demands considerable radiological expertise and is prone to inter-observer variability. The rapid proliferation of artificial intelligence in medical imaging has opened new avenues for automating and augmenting diagnostic workflows, enabling faster, more consistent analysis of complex radiological data. Despite promising advances, most existing AI-based diagnostic systems function as opaque black-box models, producing predictions without interpretable reasoning — a critical barrier to clinical adoption. LungInsight was developed to bridge this gap by building a transparent, multimodal AI framework that combines CT imaging with structured clinical data to deliver both accurate and explainable lung cancer predictions.

1.2 Purpose of the Project

The primary objective of LungInsight is to develop a multimodal deep learning system capable of predicting the probability of lung cancer malignancy and classifying disease stage from CT scan images and clinical patient records. The project further aims to integrate Explainable AI techniques — specifically Grad-CAM for imaging and SHAP for clinical features — to ensure that model predictions are interpretable and clinically trustworthy. By delivering a Streamlit-based decision support dashboard, LungInsight seeks to provide clinicians with an accessible, real-time tool to support early diagnosis and evidence-based decision-making.

1.3 Scope of the Project

This project encompasses the design, development, and evaluation of a multimodal lung cancer prediction pipeline using the LIDC-IDRI CT dataset and the National Lung Screening Trial clinical records. The scope includes CT image preprocessing, clinical data preparation, model architecture design using DenseNet-based and MLP-based encoders with late fusion, explainability integration via Grad-CAM and SHAP, and comprehensive performance evaluation. The project does not extend to real-time integration with hospital information systems, regulatory approval for clinical deployment, or the diagnosis of lung conditions other than malignancy and stage classification.

1.4 Significance of the Project

Lung cancer’s high mortality rate is closely linked to the absence of reliable, accessible, and explainable early diagnostic tools, particularly in settings with limited access to specialist radiologists. LungInsight addresses this gap by combining the predictive power of deep learning with the transparency of explainable AI, enabling clinicians to not only receive a diagnosis but also understand the imaging regions and clinical factors driving each prediction. The system has direct implications for SDG 3 (Good Health and Well-being) by improving survival outcomes through early detection, SDG 10 (Reduced Inequalities) by reducing dependence on specialist infrastructure, and SDG 9 (Industry, Innovation and Infrastructure) by advancing intelligent healthcare technology.

1.5 System Architecture

LungInsight adopts a multimodal late-fusion architecture comprising two parallel encoding branches. The imaging branch employs a DenseNet-based convolutional neural network pre-trained for grayscale CT slice analysis to extract spatial features from pulmonary nodule regions. The clinical branch uses a fully connected multi-layer perceptron to encode tabular patient data — including age, sex, smoking history, emphysema score, and lung function — into a dense embedding. The outputs of both branches are concatenated and passed through fusion dense layers to produce a final malignancy probability and stage classification. Explainability is overlaid through Grad-CAM heatmaps on CT slices and SHAP value attribution on clinical features. The complete inference pipeline is exposed via a Streamlit dashboard for

clinical use.

1.6 Problem Statement

Early-stage lung tumours are characteristically small and difficult to distinguish from benign nodules in CT scans, making manual diagnosis both time-intensive and susceptible to error. Existing AI diagnostic models, while achieving high classification accuracy, largely fail to provide interpretable reasoning for their predictions — making it difficult for clinicians to validate or trust AI-generated outputs. Furthermore, most systems rely solely on imaging data, disregarding the wealth of diagnostic signal present in structured clinical records. LungInsight addresses these limitations by proposing a unified, explainable, and multimodal prediction framework that is both clinically accurate and transparent.

1.7 Objectives

1. Develop a multimodal lung cancer prediction model using a DenseNet-based CT image encoder and an MLP-based clinical feature encoder with late fusion, to jointly predict malignancy probability and disease stage.
2. Integrate Explainable AI techniques — Grad-CAM for CT image localisation and SHAP for clinical feature attribution — to produce interpretable, patient-specific diagnostic explanations.
3. Evaluate the framework comprehensively using accuracy, ROC-AUC, F1-score, PR-AUC, Brier score, calibration curves, and decision curve analysis, with ablation studies across imaging-only, clinical-only, and fused model configurations.
4. Deploy a Streamlit-based clinical decision support dashboard enabling clinicians to upload CT scans and patient records, run inference, and review explainable risk assessments in real time.

1.8 Applications

LungInsight has broad applicability across clinical and research settings. In hospital radiology departments, it can serve as a second-opinion tool, flagging high-risk CT scans for priority review. In under-resourced or rural healthcare settings where specialist radiologists are scarce,

the system can extend diagnostic capability to general practitioners through its intuitive dashboard interface. The SHAP-based clinical explainability module is additionally valuable for oncology research, enabling identification of the most diagnostically significant patient-level risk factors. The framework’s modular architecture also allows adaptation to other thoracic malignancies or imaging modalities with minimal modification.

1.9 Structure of the Report

This report is organised as follows: Chapter 2 presents an extended literature review of AI-based lung cancer detection methods, summarising key algorithms, datasets, results, and identified research gaps; Chapter 3 details the proposed methodology, including data acquisition, preprocessing pipelines, model architecture, training protocol, and explainability implementation; Chapter 4 describes the experimental setup and implementation environment; Chapter 5 presents the results and quantitative evaluation of the LungInsight framework; and Chapter 6 offers conclusions and directions for future work.

Table 1.1: Report Organisation

Chapter No.	Content	Objective
1	Introduction	—
2	Literature Review of AI-Based Lung Cancer Detection	1
3	Proposed Methodology: Multimodal Fusion & XAI Pipeline	1, 2
4	Implementation & Experimental Setup	3
5	Results & Evaluation	3
6	Conclusion & Future Work	4

Chapter 6 Conclusion summarises the key findings of the LungInsight project, reflects on the effectiveness of the multimodal explainable AI approach for lung cancer diagnosis, and outlines directions for future research including multi-centre validation, integration with electronic health records, and extension to three-dimensional CT volume analysis.

Chapter 2

Literature Review

The application of artificial intelligence to lung cancer detection and diagnosis has grown substantially in recent years, driven by the clinical urgency of early detection and the availability of large annotated CT imaging datasets. This chapter reviews fifteen relevant works spanning CT-based deep learning, explainable AI, multimodal fusion, and pulmonary nodule detection, organised thematically to contextualise the design decisions underlying LungInsight.

2.1 CNN-Based Detection from CT Imaging

The most extensively explored paradigm in lung cancer AI involves training convolutional neural networks directly on CT scan images. Hammad et al. [?] proposed a custom CNN augmented with Grad-CAM explainability for lung cancer detection from CT images, achieving a classification accuracy of approximately 95% with visual localisation of tumour regions. While the model demonstrated strong discriminative performance, it was trained on a single dataset and showed limited generalisability across different scanner types and patient populations, with no integration into clinical workflows or longitudinal outcome tracking.

Murthy et al. [?] pursued a real-time diagnostic objective through an optimised CNN architecture, achieving fast inference suitable for clinical deployment. However, the study did not report standardised performance metrics or compare results against radiologist baselines, making reproducibility and clinical benchmarking difficult. Similarly, Prathima et al. [?] conducted a comprehensive comparative study of CNN and ensemble deep learning models on CT scans, reporting accuracies in the range of 93–97% with balanced sensitivity and specificity. Despite the robustness of the comparative design, the study's high computational cost and absence of early-stage focus limit its practical clinical applicability.

Achyutha Sree et al. [?] proposed a multi-objective deep learning pipeline on the LIDC-

IDRI low-dose CT dataset, achieving accuracies above 90% with improved PSNR and SSIM metrics and reduced false positives. The work highlighted real-time feasibility challenges and limited dataset diversity as key barriers to deployment. Gondkar et al. [?] introduced a hybrid CNN-GRU architecture combining spatial and sequential learning on CT scans, reporting accuracies of 92–95%. However, the use of GRU layers for inherently static image data was identified as architecturally redundant, and the small dataset size constrained the generalisability of results.

2.2 Explainable AI in Lung Cancer Diagnosis

Recognising that black-box predictions are insufficient for clinical trust, several works have incorporated explainability into their diagnostic pipelines. Alqahtani et al. [?] applied the EfficientNet architecture with Grad-CAM visualisation for automated lung cancer stage classification, achieving accurate stage prediction accompanied by interpretable heatmaps. The model was, however, restricted to three output classes (normal, malignant, benign), with insufficient handling of early-stage or rare subtypes and no evaluation of multimodal data fusion.

Bouamrane et al. [?] addressed model robustness by combining curriculum learning, deep CNNs, and explainable AI across multiple CT datasets, demonstrating improved cross-dataset performance and generalisation. The approach remained limited to CT imaging alone, with scalability challenged by heterogeneous preprocessing requirements and no assessment of false-positive rates across diverse demographic groups.

Zheng et al. [?] presented an interpretable computer-aided diagnosis system integrating radiology images with clinical data for malignancy evaluation, with a clear emphasis on clinical usability. While interpretability was central to the design, the work predates current foundation model developments, relies on a single dataset, and does not account for comorbidities or three-dimensional CT dynamics.

Chaudhari and Gudadhe [?] conducted a survey of AI methods applied to lung cancer diagnosis and treatment, providing a broad overview of machine learning and deep learning trends in the domain. The work was, however, non-experimental in nature, offering no quantitative benchmarking and no guidance on deployment in low-resource settings — a gap the authors themselves identified as needing urgent attention.

2.3 Multimodal Fusion Approaches

A growing body of work has sought to move beyond single-modality imaging by incorporating structured clinical data alongside CT scans. Oncu and Ciftci [?]] proposed a multimodal AI framework integrating CNN-extracted image features with ANN-processed clinical data, demonstrating improved diagnostic accuracy over imaging-only baselines. The study did not specify the datasets used or validation metrics applied, and gaps in clinical data standardisation and handling of missing modalities remain unresolved.

Yamada et al. [?]] explored multimodal deep learning by combining low-dose CT scans with structured clinical information for the diagnosis of mediastinal tumours, demonstrating the radiation-reducing benefit of low-dose imaging. The work was limited to mediastinal rather than primary lung parenchymal pathology, and the small preliminary cohort introduced bias risks with no long-term outcome data.

Niu et al. [?]] introduced a medical multimodal multitask foundation model for lung cancer screening that handles both imaging and clinical data at scale. While the architecture demonstrated strong performance across multiple screening tasks, the model remained focused on screening rather than staging or treatment recommendation, and its dependency on high-quality annotations constrained applicability in under-resourced settings. Interpretability for non-expert clinical users was also identified as a significant limitation.

2.4 Pulmonary Nodule Detection

Nodule-level detection forms a foundational component of any lung cancer diagnostic pipeline. Salhi et al. [?]] applied AI-based detection and classification models to the widely used LIDC-IDRI CT dataset, achieving improved pulmonary nodule detection accuracy. The study relied on legacy LIDC-IDRI annotations without incorporating more recent expert consensus annotations, and did not address nodule malignancy risk scoring, which is clinically essential for treatment decisions.

Koyyada et al. [?]] proposed a CNN combined with object detection for nodule identification in chest X-rays rather than CT scans, reporting accuracy in the range of 85–90% with IoU-based localisation metrics. While the approach offers a low-cost, accessible alternative to CT-based methods, X-ray modality inherently limits sensitivity for small nodules and introduces higher false-negative rates compared to CT imaging.

2.5 Survey of NSCLC Detection Methods

Pawar et al. [?] reviewed multiple datasets and AI methods applied to non-small cell lung cancer detection, reporting accuracy figures up to 97% across surveyed studies. As a review paper without experimental validation, the work provided broad directional insights but lacked deployment guidance and real-time applicability discussion — consistent with the broader pattern of a field rich in benchmarked models but sparse in clinically deployed systems.

2.6 Research Gaps and Positioning of LungInsight

Across the reviewed literature, several consistent gaps emerge. First, the majority of high-performing models rely exclusively on CT imaging, discarding the complementary diagnostic signal available in structured clinical records such as smoking history, age, emphysema score, and lung function. Second, explainability — where present — is typically limited to imaging heatmaps via Grad-CAM, with few systems offering feature-level attribution for clinical variables. Third, most works are evaluated on a single dataset, limiting generalisability across populations and scanner types. Finally, end-to-end clinical deployment with an accessible interface for non-specialist practitioners remains largely unaddressed.

LungInsight addresses these gaps by developing a multimodal late-fusion architecture that jointly encodes CT imaging through a DenseNet-based convolutional branch and clinical tabular data through an MLP-based encoder. Explainability is applied at both modalities: Grad-CAM localises suspicious regions in CT slices, while SHAP values quantify the contribution of individual clinical features to each patient-level prediction. The system is evaluated using a comprehensive suite of metrics — including ROC-AUC, F1-score, PR-AUC, Brier score, and calibration curves — and is delivered through a Streamlit dashboard that enables real-time clinical use without specialist AI expertise.

Table 2.1: Summary of Reviewed Literature

SI No.	Author(s)	Year	Method	Key Result	Key Gap
1	Hammad et al.	2025	Custom CNN + Grad-CAM	~95% accuracy	Single dataset; no clinical integration
2	Alqahtani et al.	2025	EfficientNet + Grad-CAM	Accurate stage classification	3-class only; no multimodal fusion
3	Chaudhari & Gudadhe	2025	Survey of ML/DL methods	Overview of AI applications	No quantitative benchmarking
4	Murthy et al.	2025	Optimised CNN	Fast inference	No standard metrics or radiologist baseline
5	Bouamrane et al.	2024	CNN + Curriculum Learning + XAI	Improved cross-dataset robustness	CT only; no multimodal inputs
6	Niu et al.	2025	Multimodal foundation model	Strong screening performance	Screening only; limited interpretability
7	Zheng et al.	2021	Interpretable CAD	Improved malignancy prediction	Outdated; single dataset
8	Oncu & Ciftci	2025	CNN + ANN (multimodal)	Improved diagnostic accuracy	Unspecified datasets; missing modality gaps
9	Salhi et al.	2026	AI on LIDC-IDRI	High nodule detection accuracy	Legacy annotations; no risk scoring
10	Yamada et al.	2024	Multimodal DL (low-dose CT)	Promising mediastinal results	Mediastinal scope only; small cohort
11	Achyutha Sree et al.	2025	Multi-objective DL on LDCT	>90% accuracy	High complexity; limited dataset diversity
12	Gondkar et al.	2024	CNN + GRU	92–95% accuracy	GRU redundant for static images
13	Prathima et al.	2025	CNN / Ensemble	93–97% accuracy	High cost; no early-stage focus
14	Koyyada et al.	2023	CNN + Object Detection (X-ray)	85–90% accuracy	Lower than CT; high false negatives
15	Pawar et al.	2025	Survey (NSCLC)	Up to 97% across studies	No experimental validation

Chapter 3

Design Methodology

3.1 Introduction

The design methodology for *LungInsight* is grounded in an iterative, modular development approach that mirrors the incremental refinement typical of applied deep learning research. Given the clinical sensitivity of the problem—early-stage lung cancer detection from computed tomography (CT) scans and structured patient records—the framework was architected to prioritise prediction accuracy, model explainability, and clinical deployability in equal measure. Each phase of the pipeline, from raw data ingestion through multimodal fusion to interactive clinical decision support, was designed and validated independently before integration, ensuring that failures could be localised and corrected without cascading impact on the broader system.

This methodology was selected over a purely waterfall approach because the problem formulation itself evolved during the literature survey phase. Insights drawn from ten surveyed works—spanning custom convolutional architectures, curriculum learning, multimodal foundation models, and hybrid CNN–ANN frameworks—revealed that no single prior approach simultaneously addressed the three identified gaps: multimodal fusion of CT imaging with structured clinical data, quantitative explainability of both imaging and tabular predictions, and comprehensive clinical-grade evaluation. Consequently, the development process required iterative scope refinement guided by emerging domain knowledge.

3.2 Project Requirements

3.2.1 Functional Requirements

The functional requirements of LungInsight were identified through a combination of structured literature analysis and clinical validation. The literature survey (Entries 1–15 in the extended review) consistently flagged the absence of integrated multimodal pipelines and interpretable outputs as the principal gaps in existing work. A pre-project consultation with a clinical oncologist further confirmed that explainability is not merely a research aspiration but a practical prerequisite for physician adoption of AI-assisted diagnosis tools.

The resulting functional requirements are as follows.

1. **Multimodal Input Support:** The system must accept both CT imaging volumes (DICOM/NIfTI) and structured clinical feature vectors (age, sex, smoking history, emphysema score, lung function, family history) as inputs.
2. **Malignancy Probability Estimation:** The model must output a continuous malignancy probability score for each input case, enabling risk stratification.
3. **Disease Stage Prediction:** Beyond binary classification, the system must predict the stage of lung cancer to support treatment planning decisions.
4. **Imaging Explainability:** The system must generate Gradient-weighted Class Activation Mapping (Grad-CAM) heatmaps localising suspicious regions within CT slices.
5. **Clinical Feature Explainability:** The system must compute SHAP (SHapley Additive exPlanations) values to quantify the contribution of each clinical variable to the prediction, both globally and at the individual patient level.
6. **Interactive Clinical Interface:** A web-based dashboard must allow clinicians to upload CT scans and clinical data, trigger inference, and view risk scores, heatmaps, and SHAP plots without requiring programmatic interaction.

3.2.2 Non-Functional Requirements

- **Reproducibility:** All dataset splits, preprocessing steps, random seeds, and hyperparameters must be logged to enable independent replication of results.
- **Calibration:** Model confidence scores must be calibrated such that a predicted probability of p reflects a true empirical frequency of p across the test population.

-
- **Interpretability for Non-Experts:** Explanations must be rendered in a form accessible to clinical users who are not specialists in machine learning.
 - **Modularity:** Each sub-component (imaging encoder, clinical encoder, fusion head, explainability module, dashboard) must be independently replaceable or upgradeable.

3.3 System Architecture

The high-level architecture of LungInsight is organised into four layers: the *data layer*, the *modelling layer*, the *explainability layer*, and the *presentation layer*.

The **data layer** handles acquisition, standardisation, and preprocessing of both the LIDC-IDRI CT corpus and the National Lung Screening Trial (NLST) clinical dataset. Patient-level alignment between the two sources produces coherent multimodal training instances. Preprocessing sub-modules for imaging (lung windowing, axial slicing, augmentation) and clinical data (imputation, categorical encoding, feature scaling) operate independently and feed into a unified data loader.

The **modelling layer** implements a late-fusion multimodal architecture. An imaging branch, based on a DenseNet backbone adapted for single-channel (grayscale) CT slices, extracts high-dimensional spatial feature representations. A parallel clinical branch, implemented as a Multi-Layer Perceptron (MLP), encodes structured tabular inputs into dense embeddings. A fusion head concatenates both representations and passes them through fully connected layers to produce a scalar malignancy probability and a stage classification logit.

The **explainability layer** operates post-inference. Grad-CAM hooks attached to the final convolutional block of the imaging branch generate spatial attention heatmaps. SHAP TreeExplainer (or KernelExplainer for the MLP branch) computes additive feature attributions for the clinical inputs.

The **presentation layer** is a Streamlit-based dashboard that provides an end-to-end interactive interface, integrating file upload, model inference, heatmap rendering, SHAP waterfall plots, and a clinician-facing risk summary.

3.4 Detailed Design

3.4.1 Data Acquisition and Integration

Two primary data sources are employed. The **LIDC-IDRI** dataset provides thoracic CT cases with expert-consensus annotations for lung nodule localisation and malignancy ratings (scored 1–5 by up to four radiologists). The **NLST Clinical Dataset** contributes longitudinal participant records encompassing demographic attributes (age, sex), behavioural risk factors (pack-year smoking history), physiological measurements (emphysema score, lung function indices), family history of lung cancer, and ground-truth diagnostic outcomes.

Integration proceeds via patient-level matching on shared demographic identifiers, producing multimodal tuples of the form $\langle \text{CT volume, clinical vector, label} \rangle$. The final cohort is partitioned into training, validation, and test splits with strict no-patient-overlap constraints to prevent data leakage.

3.4.2 Data Preprocessing Pipeline

CT Imaging: Raw DICOM files are converted to a standardised volumetric format. Lung windowing (window level: -600 HU; window width: 1500 HU) normalises the Hounsfield Unit range to the clinically relevant interval. Nodule-centred crops of fixed spatial extent are extracted, resampled to a uniform voxel spacing, and sliced axially. Online augmentation during training applies random rotations, horizontal flips, and brightness jitter to improve generalisation.

Clinical Data: Continuous features (age, pack-years, lung function metrics) are standardised to zero mean and unit variance. Categorical features (sex, family history, emphysema grade) are one-hot encoded. Missing values are imputed using median substitution for continuous variables and mode substitution for categoricals. The resulting fixed-length vector serves as the input to the clinical branch.

3.4.3 Model Architecture

Imaging Branch: A pretrained DenseNet (adapted for single-channel input by replicating the first convolutional layer weights across the channel dimension) processes axial CT slices and outputs a spatial feature map. Global average pooling collapses this map into a 1-D embedding vector, $\mathbf{f}_{\text{img}} \in \mathbb{R}^{d_{\text{img}}}$.

Clinical Branch: The clinical feature vector passes through a three-layer MLP with batch

normalisation and dropout regularisation at each hidden layer, producing an embedding $\mathbf{f}_{\text{clin}} \in \mathbb{R}^{d_{\text{clin}}}$.

Fusion Head: The two embeddings are concatenated, $\mathbf{f}_{\text{fused}} = [\mathbf{f}_{\text{img}} || \mathbf{f}_{\text{clin}}]$, and passed through two fully connected layers with ReLU activations. The final layer produces a scalar malignancy probability via a sigmoid activation and a multi-class stage logit via a softmax activation.

3.4.4 Training Protocol

Training uses the AdamW optimiser with a cosine annealing learning rate schedule, initialised at 1×10^{-4} . To address the class imbalance inherent in medical imaging datasets (where benign cases substantially outnumber malignant ones), the loss function combines binary cross-entropy with focal loss, modulated by an inverse class frequency weighting scheme. Early stopping on validation Area Under the Receiver Operating Characteristic Curve (ROC-AUC) prevents overfitting. Mini-batches of CT-clinical pairs are sampled with balanced class proportions.

3.4.5 Explainability Implementation

Grad-CAM (Imaging): Forward and backward pass hooks are registered on the terminal dense block of the DenseNet imaging encoder. The gradient of the predicted class score with respect to the final convolutional feature map is globally average-pooled to yield neuron importance weights. A weighted combination of the feature maps, passed through a ReLU non-linearity, produces a coarse localisation heatmap that is bilinearly upsampled to the input CT slice resolution and overlaid as a colour-coded activation map.

SHAP Values (Clinical and Fusion): For the clinical MLP branch, KernelSHAP approximates Shapley values by sampling a background reference distribution from the training set. SHAP values are computed for each clinical feature at inference time, producing both a global feature importance ranking (mean absolute SHAP across the test cohort) and a patient-specific waterfall plot showing the signed contribution of each clinical variable to the individual prediction.

3.5 Tools and Technologies

- **Python 3.11** serves as the primary development language, selected for its rich ecosystem of scientific computing and deep learning libraries.
- **PyTorch** provides the deep learning framework for model definition, training loops, and gradient computation, including the hooks required for Grad-CAM.
- **TorchVision** supplies pretrained DenseNet and ResNet weights and standard image transformation utilities.
- **scikit-learn** supports preprocessing pipelines, calibration analysis (Platt scaling, isotonic regression), and evaluation metric computation.
- **SHAP** (Lundberg & Lee) implements the KernelSHAP and DeepSHAP explainability methods for clinical and fusion branches.
- **SimpleITK / pydicom** handle DICOM and NIfTI image I/O and volumetric resampling.
- **Streamlit** powers the interactive clinical decision support dashboard, enabling browser-based inference and explanation rendering without server-side infrastructure beyond a Python process.
- **Matplotlib / Plotly** generate static and interactive visualisations for Grad-CAM overlays, SHAP plots, calibration curves, and ROC curves.
- **Weights & Biases (W&B)** tracks all experiment configurations, training metrics, and model artefacts to ensure full reproducibility.

3.6 Development Process

Development was organised into four sequential phases, each culminating in a milestone review against the project schedule.

Phase 1 – Foundation (Milestones M1–M3): Literature survey, clinical consultation, problem formalisation, and dataset procurement. The extended literature review of fifteen works established the gap space and directly informed the functional requirements. The consultation with Dr. Bindu (oncologist) validated the clinical relevance of a CT-plus-clinical multimodal approach and calibrated the scope of explainability outputs expected in a clinical workflow.

Phase 2 – Design and Architecture (Milestones M3–M4): Detailed system design, dataset preprocessing pipeline implementation, and baseline unimodal model development (imaging-only and clinical-only branches trained independently). Ablation results from this phase informed the fusion architecture decisions.

Phase 3 – Integration and Explainability (Milestones M4–M6): Late-fusion model training, Grad-CAM and SHAP integration, and dashboard development. Unit tests validated individual components; integration tests verified end-to-end inference pipelines.

Phase 4 – Evaluation and Documentation (Milestones M7–M9): Comprehensive evaluation, ablation studies, calibration analysis, and final report preparation.

3.7 Testing and Validation

The evaluation framework was designed to assess both *predictive performance* and *clinical reliability*.

Predictive Performance Metrics: Classification accuracy, sensitivity (recall), specificity, F1-score, ROC-AUC, and Precision-Recall AUC (PR-AUC) are computed on the held-out test set. ROC-AUC and PR-AUC are prioritised over accuracy given the class imbalance in the underlying dataset.

Clinical Reliability Metrics: The Brier score quantifies probabilistic forecast accuracy. Calibration curves (reliability diagrams) and Expected Calibration Error (ECE) assess whether predicted probabilities are well-calibrated against empirical outcomes. Decision Curve Analysis (DCA) evaluates net clinical benefit across a range of treatment threshold probabilities, providing a direct measure of the model’s utility in clinical decision-making.

Ablation Studies: Three model variants—imaging-only, clinical-only, and multimodal-fused—are evaluated under identical conditions to quantify the incremental contribution of each modality.

Cross-Validation: Stratified k -fold cross-validation is applied to the training set to assess variance in model performance and reduce sensitivity to the specific training–validation split.

3.8 Challenges and Solutions

Challenge 1 – Dataset Alignment: LIDC-IDRI and NLST are independently collected cohorts without direct patient-level linkage. Constructing valid multimodal training pairs required careful demographic matching, which introduced the risk of confounding. *Solution:* A

probabilistic matching strategy based on age, sex, and smoking category was implemented, and matched pairs were reviewed for demographic consistency before inclusion.

Challenge 2 – Class Imbalance: The ratio of malignant to benign cases in the combined dataset is heavily skewed toward benign, risking a model that achieves high accuracy by trivially predicting the majority class. *Solution:* Focal loss with inverse class-frequency weighting was adopted during training, and PR-AUC (which is insensitive to true negatives) was used as the primary validation criterion.

Challenge 3 – Explainability Fidelity: Grad-CAM heatmaps generated from intermediate convolutional layers can be spatially imprecise, highlighting large regions rather than specific nodules. *Solution:* Heatmaps are anchored to the final dense block of the DenseNet encoder to maximise spatial specificity, and threshold-based segmentation masks are overlaid on the original CT slice to delineate the predicted suspicious region.

Challenge 4 – Integration of Heterogeneous Modalities: Naively concatenating raw CT feature maps with clinical vectors introduces scale disparities and can cause one modality to dominate the fusion head. *Solution:* Both branches independently learn compressed embeddings of equal dimensionality before concatenation, and the clinical branch incorporates batch normalisation to align its output distribution with that of the imaging branch.

3.9 Conclusion

The design methodology for LungInsight translates the three primary gaps identified in the literature—multimodal integration, quantitative explainability, and clinical-grade evaluation—into concrete architectural decisions and a structured development process. The modular, late-fusion pipeline preserves the individual strengths of CT imaging analysis and structured clinical data processing while enabling their joint contribution to malignancy prediction. The dual explainability strategy (Grad-CAM for imaging, SHAP for clinical features) ensures that model outputs are interpretable at both the spatial and the feature-attribution levels, addressing the trust deficit that has historically limited clinical adoption of black-box AI systems. Comprehensive evaluation against calibration and decision curve metrics, beyond standard classification scores, ensures that LungInsight’s outputs are clinically actionable rather than merely statistically impressive.

Chapter 4

Implementation

The implementation phase of LungInsight translates the proposed multimodal explainable AI framework into a fully functional lung cancer prediction system. This chapter documents the concrete steps taken to build, integrate, and validate each component of the pipeline—from environment configuration and data ingestion through model training, explainability generation, and clinical dashboard deployment.

4.1 Introduction

The implementation phase focused on realising the design of the LungInsight framework as a working system capable of predicting lung cancer malignancy probability and disease stage from CT imaging and structured clinical data. The primary objectives guiding this phase were: (i) constructing a robust, reproducible data preprocessing pipeline for heterogeneous multimodal inputs; (ii) implementing and training a DenseNet-based imaging branch fused with an MLP clinical branch; (iii) integrating Grad-CAM and SHAP to produce patient-level explanations; and (iv) deploying an interactive clinical decision-support dashboard.

4.2 Development Environment

The system was developed on a workstation running Ubuntu 22.04 LTS with an NVIDIA GPU (CUDA 11.8) for accelerated model training. Python 3.10 served as the primary language for all preprocessing, modelling, and evaluation scripts. The core deep-learning stack comprised PyTorch 2.1 and torchvision; scikit-learn was used for classical preprocessing and evaluation utilities. Explainability was implemented with the `pytorch-grad-cam` library for Grad-CAM heatmaps and the `shap` library for SHAP value computation. The clinical

dashboard was built using Streamlit 1.32. Version control was managed through Git, and the project environment was containerised using Docker to ensure reproducibility across machines. Jupyter Notebooks provided an interactive environment for exploratory data analysis, while production-grade scripts were written as standalone Python modules.

4.3 Data Acquisition and Integration

4.3.1 Primary Datasets

Two publicly available sources formed the multimodal corpus for LungInsight.

LIDC-IDRI. The Lung Image Database Consortium Image Collection (LIDC-IDRI) provided 1,018 thoracic low-dose CT cases with expert annotations from four radiologists, including per-nodule malignancy ratings on a five-point scale. Raw DICOM series were downloaded from The Cancer Imaging Archive (TCIA) and organised by patient identifier.

National Lung Screening Trial (NLST) Clinical Data. Participant records from the NLST supplied structured clinical covariates: age, sex, pack-year smoking history, emphysema score, spirometric lung function indices (FEV_1/FVC), family history of lung cancer, and trial outcome labels. NLST data were obtained under a formal data use agreement and stored locally in compliance with the agreement’s terms.

4.3.2 Patient-Level Fusion

A patient-level matching script joined CT cases to clinical records via demographic attributes (age, sex, smoking category) and nodule-level descriptors. Matched records were allocated to training, validation, and test splits in an 70/15/15 ratio using stratified sampling on malignancy label to prevent data leakage; no patient appeared across more than one split.

4.4 Data Preprocessing Pipeline

4.4.1 CT Imaging Preprocessing

Each DICOM series was converted to NIfTI format using `SimpleITK`. Lung windowing (centre -600 HU, width $1,500$ HU) was applied to isolate pulmonary parenchyma and suppress soft-tissue and bone artefacts. Voxel intensities were then clipped to $[-1000, 400]$ HU and min-max normalised to $[0, 1]$. Annotated nodule centroids were used to crop 64×64 ax-

ial patches with a 5 mm contextual margin. During training, on-the-fly augmentations were applied: random horizontal and vertical flips, $\pm 15^\circ$ rotations, and Gaussian intensity jitter ($\sigma = 0.02$), improving generalisation to unseen scanner variability.

4.4.2 Clinical Data Preprocessing

Continuous covariates (age, pack-years, FEV₁/FVC) were standardised to zero mean and unit variance using statistics computed on the training split exclusively. Categorical variables (sex, family history, emphysema grade) were one-hot encoded. Missing values, present in approximately 4% of clinical records, were imputed using median substitution for continuous features and mode substitution for categorical ones. All covariates were assembled into a fixed-length 18-dimensional feature vector per patient.

4.5 Model Architecture Implementation

4.5.1 Imaging Branch

A DenseNet-121 backbone pretrained on ImageNet was adapted for single-channel grayscale CT input by replacing the first convolutional layer (weight-averaged across RGB channels) and fine-tuned end-to-end. The final classification head was replaced by a 256-dimensional fully connected (FC) projection layer with ReLU activation and batch normalisation, producing an imaging embedding $\mathbf{z}_{\text{img}} \in \mathbb{R}^{256}$.

Challenges. The single-channel adaptation initially caused instability in the first batch normalisation layer. This was resolved by freezing the backbone for three warm-up epochs and then unfreezing all layers with a differential learning rate schedule (1×10^{-4} for the backbone and 5×10^{-4} for the projection head).

4.5.2 Clinical Branch

A three-layer Multilayer Perceptron (MLP) processed the 18-dimensional clinical vector through hidden layers of size 64 and 128, each followed by ReLU activation and dropout ($p = 0.3$), and projected the result to a 64-dimensional clinical embedding $\mathbf{z}_{\text{clin}} \in \mathbb{R}^{64}$.

4.5.3 Late Fusion and Output Head

The imaging and clinical embeddings were concatenated to form a 320-dimensional joint representation:

$$\mathbf{z}_{\text{fused}} = [\mathbf{z}_{\text{img}} \parallel \mathbf{z}_{\text{clin}}] \in \mathbb{R}^{320}.$$

Two parallel FC heads processed this representation: (i) a binary malignancy classification head (sigmoid output, malignancy probability $\hat{p}$) and (ii) a four-class disease-stage head (softmax output over stages I–IV). Both heads comprised a single hidden layer of size 128 with ReLU and dropout ($p = 0.4$) prior to the output projection.

4.6 Training Protocol

4.6.1 Loss Function

The total loss combined binary cross-entropy for malignancy prediction and focal loss for stage classification to address class imbalance (stage I samples were over-represented):

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{BCE}}(\hat{p}, y_{\text{mal}}) + \lambda_2 \mathcal{L}_{\text{Focal}}(\hat{s}, y_{\text{stage}}),$$

where $\lambda_1 = 0.7$ and $\lambda_2 = 0.3$ were determined through validation-set grid search.

4.6.2 Optimiser and Scheduling

The model was optimised with AdamW (weight decay 10^{-4}). A cosine annealing learning-rate schedule with warm restarts was applied over 60 epochs. Early stopping with a patience of 10 epochs monitored the validation ROC-AUC score. The best checkpoint was retained for evaluation.

4.6.3 Training Configuration

Mini-batches of 16 CT–clinical pairs were formed, ensuring balanced representation of malignant and benign cases within each batch via a weighted random sampler. Training converged within approximately 45 epochs on the GPU-accelerated workstation.

4.7 Explainability Module Implementation

4.7.1 Grad-CAM Heatmap Generation

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the last convolutional block of the DenseNet imaging branch. For a given CT patch, the gradient of the predicted malignancy score with respect to the feature maps of the target layer was computed and global-average-pooled to derive channel weights. The resulting weighted activation map was upsampled bilinearly to the original 64×64 patch resolution and normalised to $[0, 1]$. Heatmaps were overlaid on the CT slice using a `jet` colourmap, with warmer regions indicating features most influential for the malignancy prediction. An example generation snippet is given below.

```
from pytorch_grad_cam import GradCAM
from pytorch_grad_cam.utils.model_targets import ClassifierOutputTarget

target_layer = model.imaging_branch.features.denseblock4
cam = GradCAM(model=model, target_layers=[target_layer])
grayscale_cam = cam(input_tensor=ct_patch,
                    targets=[ClassifierOutputTarget(1)])
```

Challenges. Batch normalisation in inference mode produced inconsistent activation scales across patients. This was resolved by switching all batch normalisation layers to evaluation mode and disabling gradient checkpointing during Grad-CAM inference.

4.7.2 SHAP Value Computation

A `DeepExplainer` was instantiated using a background sample of 100 training records. SHAP values were computed for each patient's clinical vector and for the fused representation, yielding feature-level attribution scores. These were visualised as waterfall plots for individual patients and as beeswarm plots across the test cohort, highlighting the relative contributions of smoking history, age, and FEV_1/FVC as the top three predictors.

4.8 Integration of Modules

The preprocessing, imaging branch, clinical branch, fusion head, and explainability components were integrated into a single PyTorch `nn.Module` class (`LungInsightModel`)

with a unified forward pass. A dedicated inference pipeline class (`InferencePipeline`) wrapped this module and handled DICOM-to-tensor conversion, clinical vector construction, model forward pass, and XAI computation in sequence, exposing a single `predict(dicom_path, clinical_dict)` interface consumed by the Streamlit dashboard. Unit-tested adapters ensured that changes to the preprocessing or model layers did not break the downstream explainability computations.

4.9 Clinical Decision-Support Dashboard

The Streamlit dashboard provided clinicians with a three-panel interface. The *Input Panel* allowed upload of a DICOM CT series and manual entry or CSV upload of clinical covariates. The *Results Panel* displayed the predicted malignancy probability as a gauge chart, the predicted stage with confidence scores, and the Grad-CAM heatmap overlaid on the most informative axial slice. The *Explanation Panel* rendered the SHAP waterfall plot for the current patient alongside reference beeswarm plots from the test cohort, enabling clinicians to contextualise individual predictions against population-level feature importances.

4.10 Testing and Debugging

Unit tests were written for each preprocessing function, model submodule, and XAI utility using `pytest`. Integration tests verified the end-to-end pipeline from DICOM ingestion to dashboard rendering with synthetic patient records. Key bugs identified and resolved included: an off-by-one indexing error in the axial patch cropping routine that caused nodule centroids to be misaligned with the output patches; a device-placement inconsistency where clinical embeddings were inadvertently left on CPU while imaging embeddings were on GPU, causing a runtime error during the fusion concatenation; and a memory leak in the Grad-CAM loop arising from retained computation graphs, resolved by wrapping inference in a `torch.no_grad()` context.

4.11 Performance Optimisation

CT patch loading accounted for approximately 65% of per-batch wall-clock time during initial training. Patches were pre-extracted from raw DICOMs and cached as compressed NumPy arrays (`.npz`) on an NVMe drive, reducing data-loading bottleneck by roughly 58%. A

`DataLoader` worker count of four with `pin_memory=True` was configured to maximise GPU utilisation. Mixed-precision training with `torch.cuda.amp` reduced GPU memory consumption by approximately 35%, allowing the batch size to be increased from 8 to 16 without exceeding VRAM limits. For dashboard responsiveness, model weights were exported to TorchScript and SHAP background samples were precomputed and cached on application start, reducing per-prediction latency to under two seconds.

4.12 Challenges and Solutions

Class Imbalance. Malignant nodules comprised only 32% of the training set. Weighted random sampling in the `DataLoader` and the focal loss term in the objective jointly addressed this imbalance, reducing the gap between sensitivity and specificity from 21 percentage points (baseline BCE only) to under 7 percentage points in the best validation run.

Multimodal Alignment. Aligning LIDC-IDRI CT cases with NLST clinical records by demographics alone introduced ambiguity for patients with identical age, sex, and smoking profiles. A probabilistic matching score incorporating nodule size and emphysema grade was introduced to rank candidate matches; cases with match confidence below 0.75 were discarded, reducing the usable paired cohort from 1,018 to 847 cases.

Explainability Faithfulness. Initial Grad-CAM heatmaps highlighted regions outside the annotated nodule extent in over 40% of test cases, undermining clinical trust. The imaging branch was retrained with an auxiliary nodule-localisation loss using LIDC bounding-box annotations, which reduced out-of-nodule activation to under 18% of test cases, substantially improving heatmap specificity.

Chapter 5

Results and Discussions

5.1 Introduction

This chapter explains the results of the proposed Explainable Artificial Intelligence (XAI) system for detecting lung cancer using CT scan images and patient clinical data. The main aim of this project was to study how deep learning and explainable AI methods can work together to improve both the accuracy of diagnosis and the understanding of how predictions are made. This chapter discusses the results obtained from the model, compares the system with existing methods, and explains the strengths and weaknesses of the proposed approach in a simple and clear way.

5.2 Presentation of Results

The proposed XAI-based lung cancer diagnosis system was trained and tested using CT scan images along with patient information such as age, smoking history, tumor size and medical background. The model showed very good performance in identifying whether lung nodules were cancerous or non-cancerous. The performance results of the model are shown below:

Table 5.1: Model Performance Metrics

Metric	Value
Accuracy	94.2%
Precision	92.8%
Recall	93.5%
F1-Score	93.1%
AUC Score	0.96

The Grad-CAM method highlighted the important areas in the CT scan images that influ-

enced the prediction made by the model. This helped users and doctors understand why the system predicted a particular result. The SHAP method explained how clinical details such as smoking history, age and tumor size affected the final diagnosis.

The results also showed that using both CT scan images and patient clinical data together improved the prediction performance compared to using only CT scan images. The combination of image data and clinical information helped the system make more accurate and reliable predictions.

5.3 Analysis of Results

The experimental results show that the proposed model works effectively for lung cancer diagnosis with high accuracy and reliability. The convolutional neural network (CNN) successfully learned important patterns and features from the CT scan images. At the same time, the patient clinical data helped the system better understand the medical condition of each patient. The explainability methods increased the transparency of the system by showing which image regions and clinical features were responsible for the prediction. This is important because doctors and healthcare professionals need to understand how AI systems make decisions before using them in real medical situations. The explainable AI methods helped confirm that the model focused on medically meaningful features instead of unrelated information. The recall value of 93.5% shows that the system can correctly identify most patients who have lung cancer. This is very important because missing cancer-positive cases can delay treatment and increase health risks. The high precision value also shows that the model produces fewer false positive results, which can help reduce unnecessary medical tests, treatments, and patient stress. Overall, the results show that the proposed XAI model is accurate, reliable, and more understandable compared to traditional black-box AI systems.

5.4 Comparison with Existing Work

The proposed XAI-based lung cancer diagnosis system was compared with other existing methods used for lung cancer detection. Traditional machine learning models usually provide lower accuracy and limited explanation of predictions. Black-box deep learning models can provide good accuracy, but they do not clearly explain how decisions are made. In medical applications, this lack of explanation can reduce trust among doctors and healthcare professionals.

The proposed XAI model not only achieved better accuracy but also provided visual and

Table 5.2: Comparison of Methods based on Accuracy and Explainability

Method	Accuracy	Explainability
Traditional Machine Learning Models	82–88%	Limited
CNN-based Black Box Models	90–92%	No
Proposed XAI Model	94.2%	Yes

feature-level explanations for predictions. The Grad-CAM method highlighted the suspicious regions in CT scan images, while SHAP explained the effect of clinical features on the final result. These explanations make the system easier to understand and more useful in hospitals and healthcare environments.

The use of both CT scan images and patient clinical data also improved the performance of the system when compared to models that use only image data.

5.5 Discussion of Key Findings

The study produced several important findings related to lung cancer diagnosis using explainable AI methods.

- Combining CT scan images with patient clinical data improved the accuracy of lung cancer prediction.
- Explainable AI methods such as Grad-CAM and SHAP improved the transparency and trustworthiness of the diagnosis system.
- The model successfully identified medically important regions in CT scan images related to cancerous lung nodules.
- Clinical features such as smoking history, age, and tumor details played an important role in the prediction process.
- The proposed system can support radiologists and doctors by assisting them in diagnosis and reducing their workload.

These findings show that explainable AI systems have strong potential in healthcare applications. The system not only provides accurate predictions but also helps doctors understand the reasons behind the predictions. This can improve trust, reliability, and acceptance of AI systems in medical diagnosis.

5.6 Limitations of the Study

Although the proposed system achieved very good results, there are still some limitations in the study.

- The model was trained using a limited dataset, which may affect its performance when used with larger and more diverse patient groups.
- Differences in CT scan quality, image resolution, and hospital imaging methods may affect the accuracy of the model.
- Clinical data may not always be available or complete in all hospitals and healthcare systems.
- Explainability methods provide supportive information, but they may not fully explain the complete internal working process of deep learning models.
- Real-time use of the system in hospitals requires more testing, validation, approval, and integration with existing healthcare systems.

Future research can overcome these limitations by using larger datasets, improving the robustness of the model, and conducting real-world testing in clinical environments.

Chapter 6

Conclusions and Future Enhancement

6.1 Conclusion

This project presented an Explainable Artificial Intelligence (XAI) system for lung cancer diagnosis using CT scan images and patient clinical data. The proposed system combined deep learning methods with explainable AI techniques to provide accurate and understandable predictions.

The experimental results showed that combining CT scan images with patient clinical information improved the performance of the diagnosis system. The use of Grad-CAM and SHAP methods increased transparency by identifying important image regions and clinical features that influenced the prediction.

The system achieved high accuracy, precision, and recall values, showing that it can become a useful support tool for radiologists and healthcare professionals. By providing both predictions and explanations, the proposed model helps improve trust and confidence in AI-based medical diagnosis systems.

Overall, the study demonstrates that explainable AI can play an important role in improving healthcare diagnosis by making AI systems more accurate, transparent, and reliable.

6.2 Future Enhancement

Several improvements can be made in future work to further improve the proposed system:

- Larger and more diverse datasets can be used to improve model accuracy and performance.
- Cloud-based real-time diagnosis systems can be developed for hospital use.

-
- Advanced transformer-based deep learning models can be applied for better feature extraction and improved predictions.
 - Explainability methods can be improved to provide more detailed medical explanations.
 - A simple and user-friendly interface can be developed for doctors, radiologists, and healthcare staff.
 - The system can be extended to detect other lung diseases such as pneumonia, tuberculosis, and pulmonary fibrosis.
 - Federated learning methods can be used to protect patient privacy while training models using data from multiple hospitals.
 - Real-world clinical testing and validation can be performed before large-scale practical deployment in healthcare environments.

Appendix A

Supplementary Dataset and Preprocessing Details

This appendix provides the complete dataset specifications, patient-matching methodology, and preprocessing steps used to support the LungInsight pipeline.

A.1 Data Sources

The imaging data were obtained from the LIDC-IDRI dataset, a publicly available collection of thoracic CT scans with multi-reader nodule annotations. Structured clinical covariates were sourced from the National Lung Screening Trial (NLST) clinical data under the applicable data-use agreement.

A.1.1 LIDC-IDRI

LIDC-IDRI contains 1,018 chest CT cases with expert annotations from four radiologists. Each nodule is annotated with bounding boxes, segmentation contours, and malignancy likelihood ratings on a five-point scale.

A.1.2 NLST Clinical Records

NLST provides demographic, smoking history, lung-function parameters, and follow-up outcomes. For LungInsight, the selected clinical variables included age, sex, pack-year smoking history, emphysema score, FEV_1/FVC ratio, family history of lung cancer, and outcome label.

A.2 Patient-Level Matching

Patient-level fusion was performed by matching each CT case to NLST clinical records using demographic and nodule descriptors. The matching process used:

- age difference within one year,
- identical sex,
- smoking category match (current, former, never),
- agreement on reported emphysema grade or lung-function impairment,
- nodule size similarity when available.

A probabilistic confidence score was computed for each candidate pair. Pairs with confidence below 0.75 were excluded to reduce false matches and prevent label noise. After filtering, the paired cohort contained 847 matched cases.

A.3 Imaging Preprocessing

The CT preprocessing pipeline converted raw DICOM series into model-ready patches using the following steps:

1. Convert DICOM volumes to NIfTI format using SimpleITK.
2. Apply lung windowing with center -600 HU and width 1500 HU.
3. Clip voxel intensities to the range $[-1000, 400]$ HU.
4. Min-max normalize clipped values to $[0, 1]$.
5. Crop 64×64 axial patches around annotated nodule centroids with a 5 mm margin.

During training, patches were augmented on-the-fly with random horizontal and vertical flips, rotations of up to $\pm 15^\circ$, and Gaussian intensity jitter with $\sigma = 0.02$.

A.3.1 Patch Cropping Algorithm

The extracted patches were generated from the axial slice containing the nodule centroid and its immediate neighbourhood. The cropping routine used the following pseudo-code:

```
for each nodule centroid:
    x0, y0 = centroid pixel coordinates
    half = patch_size // 2
    x_min = max(0, x0 - half)
    x_max = min(width, x0 + half)
    y_min = max(0, y0 - half)
    y_max = min(height, y0 + half)
    patch = slice[y_min:y_max, x_min:x_max]
    if patch size < 64:
        pad patch to 64 x 64
```

A.4 Clinical Preprocessing

Clinical covariates were preprocessed as follows:

- Continuous variables (age, pack-years, FEV_1/FVC) were standardized to zero mean and unit variance using training-set statistics only.
- Categorical variables such as sex, family history, and emphysema grade were one-hot encoded.
- Missing values were imputed with the training-set median for continuous features and the mode for categorical features.
- The final clinical feature vector had a fixed length of 18.

A.5 Data Splits

The matched paired cohort was stratified by malignancy label into training, validation, and test splits with approximate proportions of 70%, 15%, and 15%, respectively. Stratified sampling ensured that the class distribution remained consistent across all splits and that no patient appeared in more than one partition.

Appendix B

Implementation and Experimental Details

This appendix documents the supplementary implementation details, training configuration, and software environment used by LungInsight.

B.1 Model Implementation

The LungInsight architecture consists of two parallel branches:

- A DenseNet-121 imaging branch adapted for single-channel grayscale CT input, followed by a 256-dimensional projection layer.
- A three-layer clinical multilayer perceptron that processes the 18-dimensional clinical vector into a 64-dimensional embedding.

The two embeddings are concatenated into a 320-dimensional fused representation. Two output heads operate on this representation: a binary malignancy head with sigmoid activation and a four-class stage head with softmax activation.

B.2 Training Configuration

The model was trained with the following configuration:

Optimizer	AdamW with weight decay 10^{-4}
Backbone learning rate	1×10^{-4}
Projection/head learning rate	5×10^{-4}
Batch size	16 paired CT and clinical examples
Epochs	60
Scheduler	Cosine annealing with warm restarts
Warm-up strategy	backbone frozen for first 3 epochs
Mixed precision	Enabled with <code>torch.cuda.amp</code>

Early stopping monitored validation ROC-AUC with a patience of 10 epochs. The checkpoint with the best validation ROC-AUC was retained for evaluation.

B.3 Explainability and Inference

Grad-CAM was applied to the last convolutional block of the DenseNet imaging branch. During inference, all batch-normalization layers were set to evaluation mode and the Grad-CAM computation was performed inside a `torch.no_grad()` context to avoid retaining unnecessary computation graphs.

SHAP values were computed using a `DeepExplainer` with a background set of 100 representative training clinical feature vectors. Separate attributions were generated for clinical inputs and fused representations.

B.4 Dashboard and Deployment

The clinical decision-support interface was implemented with Streamlit. The dashboard accepts either a DICOM CT series upload or a CSV file containing clinical covariates. It displays:

- predicted malignancy probability,
- predicted disease stage with confidence scores,
- Grad-CAM heatmap overlay on the most informative axial slice,
- SHAP waterfall plot for the current patient and beeswarm summary for the cohort.

Model weights were exported to TorchScript for faster dashboard inference, and SHAP background samples were precomputed on startup to reduce prediction latency.

B.5 Software Environment

The implementation used the following software stack:

- Python 3.10
- PyTorch 2.1 and torchvision
- scikit-learn
- pytorch-grad-cam
- shap
- SimpleITK
- Streamlit 1.32
- Docker for containerization

The appendix is intended to provide enough technical detail to reproduce the preprocessing, model training, and dashboard deployment described in the thesis.

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